

Problem 11

CRUD ↓ heat transfer where it is deposited,

5/6 which can decrease margins such as power to melt due to ↑ temperatures due to CRUD deposits.

CRIPs - CRUD induced power shifts due to trapped boron in CRUD deposits.
- needed the radiation source

Problem 12

Boric acid → chemical shim to control reactivity

Lithium Hydroxide → Mitigate acidification of coolant due to boric acid
∴ bring pH closer to 6.9

En injection → control radiation fields

Problem 13

• Central void formation due to very high thermal gradients that cause lenticular bubbles to migrate up this gradient to the center of the fuel.
X related

• Formation of 4 distinct regions: central void, columnar grain region, equiaxed grain region & the unchanged region

• JOG → similar to rim region in LWR, but it forms band at cladding that can retain some fission products & lead to PCMI mitigation.

• Evolution of stoichiometry → Near center goes away from stoichiometry while rim area is attaining a closer stoichiometry.

Problem 6

Barium impacts the defect motion in cladding material & can make it more brittle, also attribute irradiation hardening to this & FP buildup can $\downarrow$ K_{eff} . It $\uparrow$ failure probability & makes cladding failures more likely, as well as PCMI.

Problem 7

- Reduce steam oxidation
- Improve fuel properties
- Improve cladding properties (materials)
- Improve FP ~~retention~~ retention

Problem 8

Cr coatings on cladding to prevent corrosion from high T coolant, so corrosion is less prominent, leading to safer cladding.

Problem 9

Zr cladding begins to oxidize when exposed to high T steam, which can lead to rapid hydrogen generation & subsequent hydrogen pickup, which can lead to hydriding & then cause embrittlement.

- Wanted more

Problem 10

- PCMI: fuel swelling & clad creep
- Cladding elongation & assembly bowing: anisotropic characteristic of zirconium activated by fast flux
- DNB: Film of steam insulated pin & $\uparrow$ temps in rod

Problem 2

Rate-limiting step in aq corrosion of Zr cladding is the oxidation & diffusion of oxygen. 2/2

Problem 3

6/8

Hydrides form from the oxidation of Zr cladding in coolant, which forms hydrogen & this hydrogen is taken up by the cladding & hydrides due to low solubility in Zirconium. Hydrides can cause embrittlement, especially if radial, since this can cause through-cracking - they form near the coolant-clad interface. - wanted more

Problem 4

RIA is a reactivity initiated accident. 10/12

PUR: Control Rod Ejection Accident (CREA)

BWR: Control Rod Drop Accident (CRDA)

During an RIA, the power increases rapidly, thus the temperature rises quickly within the fuel & it could lead to PCMI or ONB or both. - wanted more

Problem 5

11/12

A LOCA is a loss of coolant accident, which can lead to rapid steam oxidation of the cladding, excess decay heat, and rod pressure ↑, which can lead to ballooning or worse, burst. - still wanted some more

cause embrittlement

It is different from an RIA b/c it happens over the course of minutes rather than seconds.

Problem 1

$T = 025^\circ\text{K}$
 $t = 500 \text{ days}$
 $t^0 = 600 \mu\text{m}$

93

$a) t^*(t) = 6.62 \times 10^{-7} \exp\left(\frac{11949}{625}\right)$
 $= 133 \text{ days} < t$

$\delta^*(\mu\text{m}) = 5.1 \exp\left(\frac{-550}{625}\right) = 2.11539$

18/18

$\delta(\mu\text{m}) = \delta^* + K_L (t - t^*)$

$K_L = 7.48 \times 10^{-6} \exp\left(\frac{-7500}{625}\right) = 0.0154$

$\Rightarrow = 2.11539 + 0.0154 (500 - 133) = 7.774 \mu\text{m}$

b)

$C_H^{\text{chd}} = \frac{2(0.14) \times (7.774 \mu\text{m}) (5.68) (0.26) (1/16)}{(t - \frac{600}{1.56}) (6.5 \frac{\text{g}}{\text{cc}})}$

X/E6

200g or 15.5064

$(t - \frac{600}{1.56}) (6.5 \frac{\text{g}}{\text{cc}})$

~~1.56 g/cc~~

$\frac{110}{9.07} = \frac{32}{9.07} = 0.26$
 $\delta = 600 \mu\text{m}$

$\frac{0.2009113}{3867.61}$

$C_H^{\text{chd}} = 51.947 \text{ wt ppm}$